Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/12/2006
PI Name:	Putcha, Lakshmi Ph.D.		
Project Title:	Microcapsule Gel Formulation of Promethazine Hydrochloride for Intranasal Administration		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:	Deceased as of September 2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	08/01/2001	**End Date:**	08/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A continuing challenge for space medical operations at NASA is the management of pathology associated with neurovestibular adaptation during space flight. A primary manifestation of this problem, particularly in the first few flight days of shuttle missions, is space motion sickness (SMS). The current treatment of choice for symptoms associated with SMS is promethazine (PMZ). Although oral tablets and rectal suppositories have been used during space flights, the intramuscular route appears to be most effective. On the other hand, intramuscular administration of drugs is an invasive procedure and PMZ causes irritation at the site of injection. A key research topic in the Smart Medical Systems area of the NSBRI 99-02 research announcement is development of novel therapeutic modalities for remote site medical operations such as space missions. In response to this initiative, the goal of the proposed research is to develop an intranasal dosage formulation of PMZ that will provide crewmembers with a non-invasive means of self-administering SMS medications. Accordingly, the following three aims will be addressed: 1) Develop a microencapsulated, pH		

	balanced gel dosage formulation and a combination form with a corticosteroid for intranasal administration of PMZ; 2) Establish the release kinetics and shelf life of the optimized dosage forms, and 3) Assess bioavailability, nasal mucosal irritability and toxicity of the selected dosage forms in rats. The proposed formulation development will focus on tailoring the release characteristics of the dosage form to optimize therapeutic index and minimize irritability at the site of administration. Once the optimal dosage form has been identified based on release kinetics and stability characteristics, bioavailability, nasal irritability and toxicity after single and multiple dose administration will be assessed in an animal model. Development of an intranasal drug delivery system for motion sickness treatment will benefit pharmacotherapeutics in space as well as on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microcapsule gel formulations are state-of-the-art dosage forms that will have a lot of potential in acute and chronic treatment modalities. Intranasal administration of drugs will offer an excellent alternative to conventional oral and parenteral routes, especially for pediatric and geriatric clinical care.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 06/23/2025)